Bluestock Mutual Fund Analytics

Capstone Project Final Report

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1. Executive Summary

The Bluestock Mutual Fund Analytics Capstone Project was designed to bridge the gap between raw financial data and executive-level Business Intelligence. Over the course of the project, a robust end-to-end data pipeline was constructed to ingest, process, and analyze complex mutual fund data encompassing 40 schemes across the Indian asset management industry.\n\nBy transitioning from disparate CSV files into a centralized SQLite Star Schema Data Warehouse, the project enabled advanced quantitative modeling and behavioral demographic analysis. Key deliverables included the extraction of manager skill (Alpha) via Ordinary Least Squares regression, the calculation of deep tail-risk metrics (Value at Risk and Conditional VaR), and the deployment of an automated Churn Prediction algorithm for retail SIP investors.\n\nThe project culminated in the development of a highly interactive, Python-based Business Intelligence Dashboard built on Streamlit, allowing stakeholders to dynamically cross-filter historical performance, track industry milestones (such as the INR 81 Lakh Crore AUM mark), and interact with a proprietary algorithmic 0-100 Fund Scorecard.

2. Data Sources & ETL Architecture

The foundation of the analytics platform was built upon 10 distinct, raw datasets spanning daily Net Asset Values (NAV), retail investor transactions, portfolio sector holdings, and benchmark indices (Nifty 50 and Nifty 100).\n\nETL Pipeline Design:\n1. Extraction: Python's Pandas library was utilized to read, clean, and standardize the raw CSV files, handling missing values through forward-filling techniques (for holiday NAV gaps) and validating data types.\n2. Transformation: Complex transformations were applied to standardize transaction types (SIP, Lumpsum, Redemption) and normalize demographic categorical variables.\n3. Loading (Star Schema): The normalized data was ingested into a SQLite database (bluestock_mf.db) using SQLAlchemy. The architecture was modeled as a Star Schema:\n - Dimension Tables: dim_fund (scheme metadata), dim_date (temporal mapping).\n - Fact Tables: fact_nav (daily pricing), fact_transactions (retail flows), fact_aum (assets), and fact_performance (aggregated metrics).\n\nThis relational structure ensured optimal querying performance for the downstream Python analytics scripts and the interactive dashboard.

3. Exploratory Data Analysis (EDA) Findings

The automated EDA pipeline revealed significant macro and micro trends within the mutual fund landscape:\n\nMacro Growth & Asset Under Management (AUM):\nThe industry exhibited massive structural growth between 2022 and 2025, culminating in an all-time high Total AUM of roughly INR 81 Lakh Crores. Market share remains highly concentrated among the top 3 Asset Management Companies (AMCs), with SBI Mutual Fund commanding a dominant position.\n\nRetail Participation & The SIP Boom:\nSystematic Investment Plans (SIPs) proved to be the primary engine of capital inflow, reaching a peak monthly volume of INR 31,002 Crores by December 2025. This retail influx pushed total investor folios to 26.12 Crores.\n\nDemographic Insights:\nGeographically, Maharashtra and Gujarat contributed the highest volume of SIP inflows. When analyzed by age group, investors aged 25-35 generated the highest transaction frequency, representing a massive new cohort entering the market. However, investors in the 45-55 age bracket maintained the highest average ticket size per transaction.

4. Advanced Performance & Risk Analytics

Transitioning from historical observation to quantitative evaluation, several sophisticated models were deployed to rank and stress-test the 40 mutual fund schemes.\n\nAlpha & Beta Extraction:\nUsing scipy.stats, an Ordinary Least Squares (OLS) regression was performed for each fund against the Nifty 100 index. This successfully decoupled Beta (market correlation) from Alpha (idiosyncratic manager skill). Several highly-rated funds were exposed as 'closet indexers' offering high Beta but zero or negative Alpha.\n\nRisk-Adjusted Metrics:\nThe Sharpe Ratio (calculated against a 6.5% risk-free rate proxy) and Sortino Ratio were computed to evaluate risk-adjusted returns. Furthermore, the 90-Day Rolling Sharpe ratio was plotted, revealing that static 3-year averages often mask periods of severe managerial underperformance during market volatility.\n\nTail Risk & Expected Shortfall:\nStandard deviation assumes a normal distribution, which fails during 'black swan' events. The pipeline calculated the Historical 95% Value at Risk (VaR) and the Conditional VaR (Expected Shortfall). Small Cap funds exhibited CVaR metrics nearly 2.5x worse than Large Cap peers, quantifying extreme tail risk.\n\nThe Composite Scorecard:\nAn algorithmic grading system (0-100) was engineered, weighting 3-Year CAGR (30%), Sharpe Ratio (25%), Alpha (20%), Expense Ratio (15% inverse), and Max Drawdown (10% inverse). This created an objective, mathematically sound ranking system for the dashboard.

5. Dashboard Implementation

To operationalize the analytics, a 4-page Business Intelligence Dashboard was developed using Streamlit, Python's industry-standard framework for data applications. This programmatic approach allowed for direct integration with the SQLite Star Schema and native Python libraries, offering superior version control and deployment capabilities compared to proprietary tools like Power BI.\n\nDashboard Architecture:\n1. Industry Overview: Displays dynamic KPI cards tracking total AUM, SIP flows, and active folios.\n2. Fund Performance: Features an interactive Risk vs. Return scatter plot and the sortable, algorithmic Fund Scorecard matrix.\n3. Investor Analytics: Visualizes demographic cohorts, geographical distribution, and transaction type splits.\n4. SIP & Market Trends: Unveils macro behavioral patterns, including a dual-axis chart mapping SIP inflows directly against the Nifty 50 trajectory to validate the 'Sticky Capital' hypothesis.

6. Limitations & Recommendations

Limitations:\n1. Time Horizon: The 3-year historical dataset (2022-2025) is insufficient for full-cycle economic stress testing, as it largely represents a singular bull market with minor corrections.\n2. Granularity: The lack of daily individual stock transaction data from the fund managers restricts the ability to perform deepest-level performance attribution (e.g., sector timing vs. stock selection).\n\nRecommendations for Future Development:\n1. Churn Intervention Deployment: The advanced analytics pipeline successfully flagged a cohort of 'at-risk' investors whose average SIP gap exceeds 35 days. This data should be immediately fed into the CRM system to trigger automated retention workflows.\n2. Retail Recommender Integration: The lightweight Python Recommendation Engine (recommender.py) built during Day 6 should be integrated into the Bluestock mobile application via REST API, providing retail investors with automated, risk-adjusted fund selections based on their risk appetite profiles.\n3. Live Data Feeds: Transition the ETL pipeline from batch CSV processing to live API webhooks (e.g., AMFI APIs) to ensure the Star Schema is updated in real-time.

Appendix 1: Supplementary Data

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